

MTH 553: Lecture # 39

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Lecture Span

1. Continuation of trajectory

Let

$$E(v, w) = \int_0^v f(z) dz + \frac{1}{2} w^2 \quad (1)$$

We showed that

$$\frac{d}{ds} E(v(s), w(s)) = c w^2(s) \geq 0 \quad (2)$$

This means that the energy increases along trajectories. We note that

$$E(0, 0) < E(1, 0) \quad (3)$$

Since

$$\int_0^1 f(v) dv > 0 \quad (4)$$

For large c , the trajectory cuts through the level sets (lines where the energy is constant) quickly. That is, the energy increases rapidly. Here, then the trajectories do not meet.

For small c , the energy will change slowly, but then the trajectory will not meet (undershot).

Therefore, there must be some in-between value of c of which these trajectories will meet.

Schrödinger's equation on $\mathbb{R}^n$

Only difference between SE and heat equation is "i". Recall, we found the fundamental solution to the heat equation by Fourier Transform.

$$u_t = \Delta u \quad (5)$$

With

$$u(x, t) = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} \exp(-|x - y|^2/4t) g(y) dy \quad (6)$$

Then Schrödinger's equation

$$iu_t + \Delta u = 0 \quad (7)$$

On $\mathbb{R}^n$. With $u(x, t) \in \mathbb{C}$, or is a complex function. Just the heat equation but with t replaced by it . Then

$$\frac{\partial u}{\partial(it)} = \Delta u \iff \frac{1}{i} \partial_t u - \Delta u = 0 \quad (8)$$

Multiply by $i^2 = -1$ to obtain SE. So then replacing t with it should obtain the fundamental solution of Schrödinger's equation.

$$u(x, t) = \frac{1}{(4\pi it)^{n/2}} \int_{\mathbb{R}^n} \exp(i|x - y|^2/4t) g(y) dy \quad (9)$$

Should solve Schrödinger's equation with $u = g(x)$ at $t = 0$. For $x \in \mathbb{R}$ and for all $t \in \mathbb{R}$ and $t \neq 0$. You can check (9) directly by differentiating through the integral. We also assume that $g \in L^1$ and $|y^2|g(y) \in L^1$. That is, g is integrable. Does $u(x, t) \rightarrow g(x)$ at $t \rightarrow 0$? Yes, using the method of stationary phases.

SE is time reversible. That is $u(x, -t)$ also solves (SE). For the Schrödinger on the bounded domain $\Omega \subset \mathbb{R}^n$ with Dirichlet boundary conditions, then the solution is written in terms of eigenfunctions $\{\phi_k\}$.

$$u(x, t) = \sum_k c_k \exp(-i\lambda_k t) \phi_k(x) \quad (10)$$